6.1 Text Classification Pipeline

The full path from raw text to a label.

These notes walk through the lecture slides. The original deck is unchanged; use **(slides)** on the course hub if you want that PDF.

1. Classification is supervised

Supervised learning means the training rows come with the answer. Those answers are **labels**.

Two jobs dominate:

- **Classification:** a discrete bucket. Spam vs ham is the running example. The label is a class.
- **Regression:** a number. Price of a car from mileage and brand. The label is continuous.

This week is classification of **text**. The input is a document or snippet. The output is a predefined category.

Algorithms you already know from a first ML course still apply once the text is a vector:

Algorithm	Typical role in this course
k-NN	Simple baseline, no training of weights
Logistic regression	Strong linear baseline on TF–IDF
Linear SVM	Same, often a bit better on high-dimensional text
Trees / forests	Nonlinear, weaker on raw sparse bags
Neural nets	Weeks 7–8 once you have embeddings

Unsupervised methods (clustering, PCA, t-SNE, LDA) have no labels. They organize or compress. They are not this week's classifier, though you may use them to **look** at the data.

The main text is *Practical Natural Language Processing*.

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2. What text classification is

You assign each document a label from a fixed set: sentiment, topic, spam, language, priority. The model learns patterns from a **labeled** collection, then labels new text.

It is the most common applied NLP job in this course. If you can take a CSV of texts and labels and ship a scored model, you can do the rest of the semester's projects.

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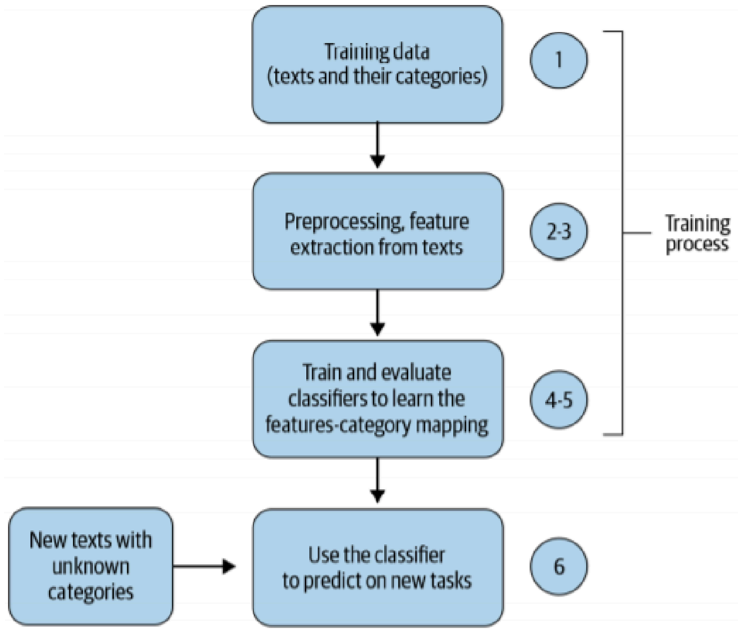
3. The pipeline

Every system has the same spine. Representation is Week 4–5. This week you put a **classifier** and an **evaluation** on the end.



In order:

1. **Collect and label.** The bottleneck is usually labels, not models.
2. **Clean and tokenize.** Lowercase, strip junk, decide what a token is.
3. **Vectorize.** BoW, TF–IDF, or embeddings. Fit the vectorizer on **train only**.
4. **Train.** Pick a model family and its hyperparameters on a validation split or inner CV (note **6.2**).
5. **Evaluate.** Accuracy if classes are balanced; precision, recall, F1 if they are not.
6. **Ship and watch.** New topics and new slang shift the distribution. A model that was fine in March can fail in September.

Skip a box and you will debug the wrong thing. A “bad SVM” is often a leak (vectorizer fit on the whole corpus) or a metric that hides a majority class.

4. Where this shows up

Application	What the label is
Content organization	Topic or desk (sports, politics)
Misinformation	Claim is supported / not, or source is reliable / not
Authorship	Which writer (or which account)
Customer support	Intent, queue, or urgency
E-commerce	Category, aspect sentiment, fake-review flag

Support and e-commerce are the pictures in lecture because the **cost of a wrong bucket** is obvious: the ticket goes to the wrong team, or the product lands in the wrong aisle. Build the pipeline so you can measure that cost, not just accuracy.

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5. What note 6.2 adds

This note is the map. Note **6.2** is the classical models and the evaluation you will use in lab: k-NN, SVM, cross-validation, parameters vs hyperparameters, precision and recall.

Do not start Week 7 until you can draw the flowchart and name a metric for an imbalanced ticket queue.

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6. Practice

1. Is topic modeling (LDA) text classification? Why or why not?
2. You fit `TfidfVectorizer` on train+test together, then split. What did you leak?

3. Pick one application in the table. Name the classes, who labels the data, and what a false positive costs.